Consider using these various techniques to help answer each of the following limits.

1. Compute the limit $\lim_{h \to 0} \frac{(2+h)^2 - 4}{h}$
2. $\lim_{u \to 2} \frac{\sqrt{u+7} - 3}{u-2}$
3. $\lim_{x \to -3^-} \frac{81 - x^4}{x+3}$

Follow-up: Does it matter that this is a limit from the left? Should this affect the answer?

4. $\lim_{x \to -1} \frac{x^2}{x^3 - x}$
5. $\lim_{t \to 0} \left(\frac{1}{t^3} - \frac{t-1}{t^4} \right)$
6. If $\frac{x^2 - 5x + 6}{x-2} \leq g(x) \leq x^2 - 3x + 1$ for all x, determine $\lim_{x \to 2} g(x)$
7. $\lim_{\theta \to 0} \frac{\sin(\theta)}{\sec(\theta)\sin(2\theta)}$
8. $\lim_{x \to 1} (x-1)^4 \cos^2\left(\frac{3+x}{x^2 - 1} \right)$

9. Consider this piecewise function $f(x)=\begin{cases} \dfrac{\sqrt{x+1}-1}{x}, & x<0 \\ \dfrac{x}{x-2}, & 0\le x<2 \\ \dfrac{x-2}{x^3-2x^2+x-2}, & x\ge 2 \end{cases}$.

a) Create a limit for $f(x)$ that equals $\frac{1}{2}$.

b) Create a limit for $f(x)$ that equals -1.

c) Create a limit for $f(x)$ that equals $\frac{1}{4}$.

d) Create a limit for $f(x)$ that DNE.

e) Create a limit for $f(x)$ that equals negative infinity.

f) Create a limit for $f(x)$ that equals $\frac{1}{101}$.

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